ENGG1300

Notes for HKU · Spring 2024

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1 Vectors

Basics

Vector can be represented by $\vec{F}$ or $\boldsymbol{F}$. The magnitude can be represented by $|F|$ or F.

Unit vectors

They are vectors with magnitude 1. $\hat{A} = \frac{\vec{A}}{|\vec{A}|}$

1.1 Coplanar vectors

Cartesian vector notation

In two dimensions, the Cartesian unit vectors $\boldsymbol{i}, \boldsymbol{j}$ are used to designate the directions of the x and y axes respectively.

$$F = F_x \boldsymbol{i} + F_y \boldsymbol{j}$$

Where $F_{x/y}$ is the x/y component of F. And to find the x/y components we can use trigonometry:

$$F_x = F \cos\theta, \quad F_y = F \sin\theta$$

Where the angle θ is the angle between F and the x-axis.

Resultant force

The resultant force F_R can be found by the sum of the components of F:

$$F_R = \sum F$$

In Cartesian form, it's the same as adding all the terms together: $F_R = (F_{x1} + F_{x2})\boldsymbol{i} + (F_{y1} + F_{y2})\boldsymbol{j}$

Orientation of vector

We always consider the angle between F & F_x. It can be found by $\theta = \tan^{-1} \frac{F_y}{F_x}$.

Magnitude of forces

The magnitude will simply be the square root of the sum of squared components of the force:

$$|F| = \sqrt{F_x^2 + F_y^2 + \dots}$$

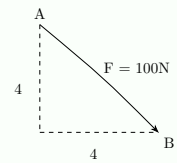
Moving vectors by their position vectors

We can move a force F to a point using its position vector r (pointing to their tail), using the following definition:

$$\vec{F} = |F| \times \hat{F} = |F| \times \frac{r}{|r|}$$

The position vectors are in Cartesian form, so the moved vectors will also be in Cartesian form.

Consider the following graph:



First, we consider r_{AB}. From the graph:

$$\vec{r_{AB}} = 4\boldsymbol{i} + 4\boldsymbol{j}$$

Then, we find the magnitude of r_{AB}:

$$|r_{AB}| = \sqrt{4^2 + 4^2} = 5.65\ldots$$

Finally, we apply our formula for F:

$$\begin{aligned}\vec{F} = |F| \times \frac{r_{AB}}{|r_{AB}|} &= 100 \times (\frac{r_x}{5.65}\boldsymbol{i} + \frac{r_y}{5.65}\boldsymbol{j}) \\ &= 100 \times (\frac{4}{5.65}\boldsymbol{i} + \frac{4}{5.65}\boldsymbol{j}) \\ &= 70.7\boldsymbol{i} + 70.7\boldsymbol{j}N\end{aligned}$$

1.2 Vectors in 3D

The concepts above can be extended to 3D simply by adding another variable to the system.

2 Moment of forces

Definition of moment

The moment of a force is a measure of its *tendency* to cause a body to rotate about a specific point. The moment about a point O, when F is applied a distance d from the point is:

$$M_O = F \times d$$

Keep in mind that *positive* moment is *anti-clockwise*.

2.1 Coplanar / 2D moment

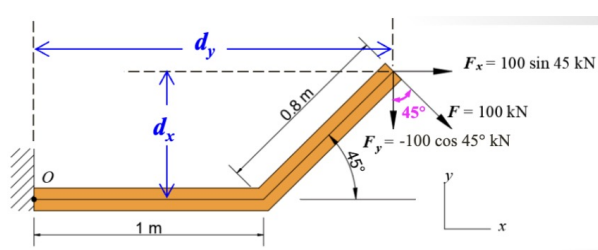
Resultant moments

The resultant moment is the **sum** of all moments present on the point, given by:

$$M_R = \sum M_O$$

2.1.1 The moment of a non-linearly attached force

One simple way is to find the *components* of the force, and sum their individual moments together. The following is a simple example:



After finding the component forces of F, we can deduce the resultant moment to be:

$$\begin{aligned} M_O &= F_x \times 0.8 \sin 45\,\text{deg} + F_y \times (1 + 0.8 \cos 45\,\text{deg}) \\ &= -150.7 kNm \end{aligned}$$

2.2 Non-coplanar / 3D moment

Moments in a 3D system

Consider position vector $\vec{r}$ drawn from O to any point on the *line of action* of F. The moment can hence be given by:

$$M_O = r \times F$$

Finding the moment via cross products Cartesian vectors

The cross product C given by A and B is:

$$A \times B = \begin{vmatrix} i & j & k \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} = C$$

The cross-product for vectors going in the *same direction* is 0. (i.e. $n\boldsymbol{k} \times m\boldsymbol{k} = 0$)

Resultant moments

The resultant moment is simply the **sum** of *couple moments* and moments of forces:

$$(M_R)_O = \sum M_O + \sum M$$

You can interpret $(M_R)_O$ as the resultant moment about point O.

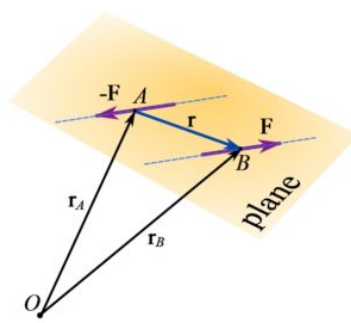
2.3 Couple moments

Couples are *two parallel forces* that have the same magnitude but have *opposite directions*, separated by a *perpendicular distance* d. The magnitude of the moment is given by:

$$M = Fd$$

Notice that there's no point mentioned so far. For couple moment, it is **always the same about any point**. Let's assume for any point O (refer to graph), the moment is:

$$\begin{aligned} M_O &= r_B \times F + r_A \times -F \\ &= (r_B - r_A) \times F \\ &= r \times F \quad \text{which is independent of } O \end{aligned}$$



Hence, we can say that couple moments are **free vectors**.

3 Axially loaded members

Axial loading refers to the application of a force along the axis of the member.

3.1 Stress and strain

Axial stress

Axial stress is the stress that is *parallel* to the cross-sectional area of the member. It is given by:

$$\sigma = \frac{F}{A} \quad (Nm^{-2})$$

Where F is the force applied, and A is the cross-sectional area of the member. Note that $1Pa = 1Nm^{-2}$.

Eccentric loading and stress

When a force is applied *off-centre* to the member, the stress at each end is given by:

$$\sigma = \frac{F}{wd} \pm \frac{6F \times e}{d \times w^2}$$

Where w is the width of the member, d is the depth of the member, and e is the eccentric distance from the centroid of the member to the point of application of the force.
The $\pm$ sign is used to denote the *maximum* and *minimum* stress on opposite sides.

Axial strain

Axial strain is the ratio of the change in length to the original length of the member. It is given by:

$$\epsilon = \frac{\Delta x}{x} \quad \text{(Ratio)}$$

Where Δx is the change in length, and L is the original length of the member.

3.2 Materials

Strength

Material strength is defined as the **maximum stress** that can be resisted by the material.

Young's Modulus

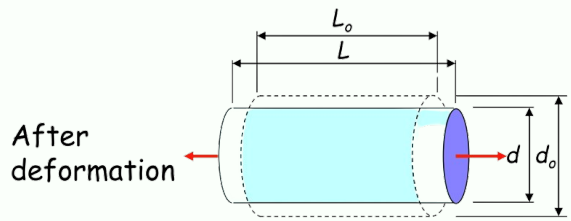
Young's modulus is the ratio of stress to strain, given by:

$$E = \frac{\sigma}{\epsilon} = \frac{Fx}{A\Delta x} \quad (Pa, Nm^{-2})$$

Poisson's ratio

Poisson's ratio is the ratio of lateral strain ϵ_l to axial strain ϵ, given by:

$$\nu = \frac{\epsilon_l}{\epsilon} \quad \text{(Ratio)}$$



The lateral strain ϵ_l is $\frac{\Delta d}{d_0}$.

3.3 Hydrostatic pressure

Hydrostatic/water pressure

The water pressure acting on any surface is *always perpendicular* to the surface, and the pressure is given by:

$$p = \rho g h \quad (Pa, Nm^{-2})$$

Where ρ is the density of water, and h is the depth of the water.
Hence, we can see that the water pressure **increases linearly with depth**.

Water pressure load on slanted surface

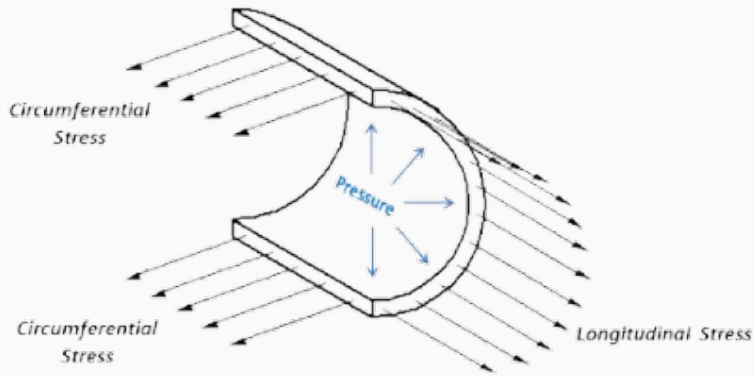
The load exerted on a slanted surface by water pressure F is given by and located at:

$$F = \frac{\rho g d w L}{2} \quad @\frac{1}{3}L \ / \ \frac{2}{3}d$$

Where d is the depth of water, w is the width of the volume, and L is the length of the surface.

Water pipe

1. **Internal pressure** refers to the pressure inside the pipe.
2. **Internal force** refers to the pressure's effect onto a side of the pipe.
3. **Internal stress** refers to the stress caused by the internal force, acting along the *circumferential direction* of the pipe. This is also called the **hoop stress**.



4 Statics

4.1 Equilibrium of rigid bodies

The 3 equations of equilibrium

A rigid body is in equilibrium if:

$$\sum F_x = 0, \quad \sum F_y = 0, \quad \sum M_O = 0$$

Where O is any point.

Static equilibrium

A system is in **static equilibrium** if it experiences no acceleration when loads are applied to it.

Free body diagrams (FBD)

Free body diagrams are diagrams that show all the forces acting on a body. They are useful in determining the forces that cause the body to be in equilibrium.

4.2 Members

A member is a *straight* and *long* structural element that is subjected to axial forces. The following are the types of members:

4.3 Static systems

Supports and reaction forces

Each support will restrict the movement of the rigid body in a certain way. If a certain degree of freedom is restricted, a reaction force will be present to counteract the force that would have caused the movement.

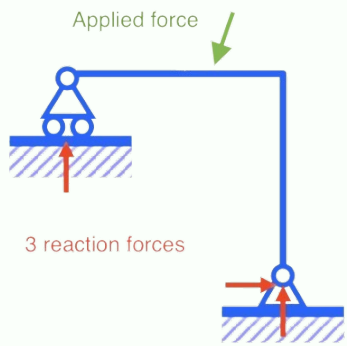
Support type	Restricted movement and reaction forces	Shape
Fixed	x, y, r (moment)	Flat
Pinned	x, y	Triangle
Roller	y	Circle

Static conditions of a system

A system's state of equilibrium can be determined by the number of restraints present:

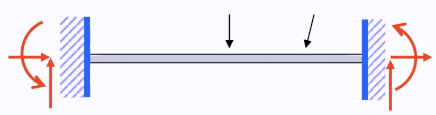
1. Insufficient restraints $\rightarrow$ **non-static system** (contains mechanisms)
2. Sufficient restraints $\rightarrow$ **static system**

If a static system has **3 reaction forces**, the system is **statically determinate**, if there are more, it is **statically indeterminate**.



Degree of indeterminacy

For a statically indeterminate system, the degree of indeterminacy (or redundancies) is simply $R - 3$ where R is the number of reaction forces.



The number of redundancies is 3.

4.4 Loading

Types of loading

1. **Concentrated load** is a force applied at a single point. (N)
2. **Distributed load** is a force applied over a length. (N/m)

Distributed load

A *distributed load* (w) is a force that is distributed over a length (l), that has the unit N/m.
The **resultant force** is the supposed area of the load. To consider the **moment** by a distributed load, we can treat the load as a *single force* acting at the **centroid** of the load.

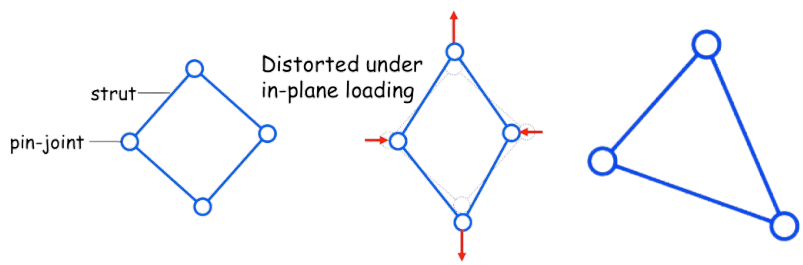
- Unifrom $\rightarrow F = wl \quad @\frac{1}{2}l$
- Triangular $\rightarrow F = \frac{1}{2}wl \quad @\frac{1}{3}l$ (from the base of the load)

5 Structural analysis

5.1 Pin-joined trusses

A **truss** is a structure that consists of *straight members* connected at their ends by *pin joints*.

A triangular truss is a statically determinate structure, but a quadrangular truss is not.

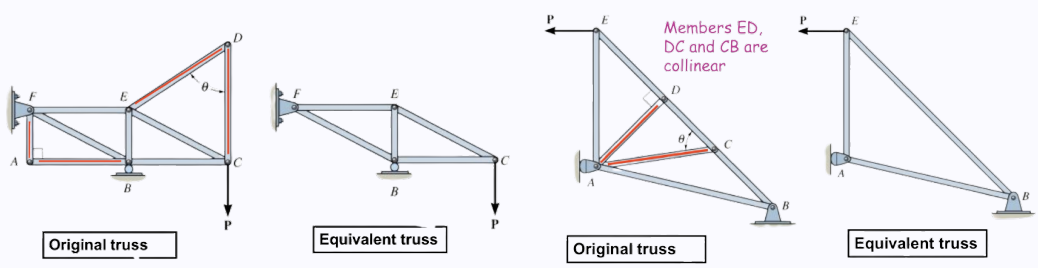


Zero-force members

The following are zero-force members given that **no external load or support reaction is applied to the joint**.

- **2 non-collinear members** of a two-member-joint
- **3rd member** of a three-member-joint, where the other two are collinear.

Removing zero-force members allows us to simplify a truss structure.



5.2 Statically determinate trusses

To construct an *internally* statically determinate truss, we begin with a triangular pin-jointed truss and then successively adding two new members with a new joint.

We say the truss is only *internally* statically determinate because the external reactions are not yet known.

5.3 Truss analysis

When we analyze a truss, we assume that forces are only applied at joints.

Internal vs. External forces

- Forces acting on the members $\rightarrow$ **Internal forces**
- Forces acting on joints $\rightarrow$ **External forces**

Sign convention of analysis

As we focus on the analysis of **internal forces acting on joints**, we always draw:

- Tension forces $\rightarrow$ **pointing away** from the joint $\{\leftarrow \circ \boxed{\rightarrow \ T \ \leftarrow} \circ \rightarrow\}$
- Compression forces $\rightarrow$ **pointing towards** the joint $\{\rightarrow \circ \boxed{\leftarrow \ C \ \rightarrow} \circ \leftarrow\}$

During analysis, we always draw forces **pointing towards** the joint $\{\rightarrow \circ \leftarrow\}$, which means we treat **compression as positive**.
We also treat **downward forces as positive** $\{+ \downarrow\}$.

5.4 Equilibrium sections

If we cut out any section of an equilibrium system, the cut will be in equilibrium. That is, the **external forces** are balanced by **internal forces**.

Therefore, to solve for the forces in a system, we simply cut out a section that we deem solvable (by practice!) and draw it's FBD. The forces includes *all external forces in the cut* as well as the *interal forces of the cut members*.

5.5 Cable anaylsis

Cable forces

Cables are **always in tension**. The tension force is always in the direction of the cable.

Some characteristics of cable structures:

- Supports are always 2 inverted pinned supports
- Applied forces all point downwards due to gravity
- Horizontal reaction forces at supports are equal in magnitude

Hence, we can use the following steps to solve for cable forces:

1. Find reaction forces by considering joint equilibrium
2. Tension in cables attached to a support can be found by $\sqrt{(R_x)^2 + (R_y)^2}$
3. Tension in cables not attached to a support can be found by $\sqrt{(R_x)^2 + (R_y - \sum y)^2}$

6 Bending forces and stress

6.1 Internal force in beams

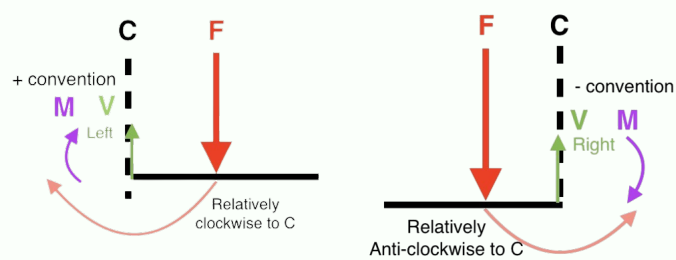
When we consider the internal forces of a beam, we take a *cross section* and consider the forces acting on it. The internal forces are:

- **Shear force** V is the force acting parallel to the cross-section.
- **Bending moment** M is the moment generated by the variation of forces acting perpendicularly to the cross-section.

Sign convention of internal forces

$\{+ \circlearrowright\}$ The idea is that if the acting force is relatively **clockwise** to the cross-section C, it is **positive**. The following is the signs of internal forces for a *positive* moment:

- **V:** $\circlearrowright$ (L $\uparrow$ R $\downarrow$)
- **R:** $\curvearrowright$ $\curvearrowleft$ (sagging moment is *positive*)



Couple moments can be treated as a rotational moment at clock-direction the arrows are pointing to.

Finding internal forces by side

To find the internal forces at cross-section C, we consider the *left* and *right* side of the section:

- **V:** $\sum F$ on either side
- **M:** $\sum M_C$ on either side about C

Note that the sums are all **signed by convention**. (e.g. A force F is positive if it points upwards on the left side.)
This also gives us the fact that $\sum F_L = \sum F_R$ and $\sum M_L = \sum M_R$, signed by convention on both sides.

6.2 Diagramming internal forces

Diagramming internal forces

The two internal force diagrams are $V(x) - x$ and $M(x) - x$ diagrams. $V(x)$ gives the shear force V at distance x from the *left side* of the beam.

Steps to follow:

1. Solve for reaction forces
2. Draw the FBD of a cut from the leftmost side to a location with distance x from the leftmost force *not analysed*
3. Analyse the FBD of the cut (solve for V and M)
4. Draw the graphs in the cuts and repeat for all cuts

Figurative steps:

1. $\rightarrow R_A, R_B ...$
2. $\{\overset{R}{\uparrow} \underbrace{=====}_{x} \overset{V\,M}{\downarrow\circlearrowleft} \dots\}$
3. $\rightarrow V, M$
4. $\rightarrow V(x), M(x)$

To analyse a distributed load w, consider the FBD of the whole system on the left, then let x be the distance from the leftmost force *analysed*:

$$\{\underset{R}{\underset{\uparrow}{=}}\underset{2}{=}\overset{F}{\underset{\downarrow}{=}}\underbrace{\overset{w\cdot x}{\underset{\downarrow}{=}}=}_{x}\overset{V\,M}{\downarrow\circlearrowleft}\dots\}$$

6.3 Bending stress

7 Centroids and moment of inertia